



CAITLIN ADAMS

SENIOR DATA SCIENTIST



AUSPACIOUS

CONTACT

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SKILLS

- Geospatial Data Science
- Remote Sensing (optical and SAR)
- Machine Learning
- Cloud Native Geospatial Metadata (STAC)
- Advanced Data Visualisation
- Technical Leadership
- Project Management
- Project Delivery
- Training and Workshop Design and Delivery

EDUCATION

- Doctor of Philosophy in Astrophysics | Swinburne University of Technology
- Bachelor of Science in Physics (First Class Honours) | University of Queensland



PROFILE

Caitlin Adams brings a background in astrophysics to the challenge of turning satellite data into insights for the planet. After a PhD using galaxy survey data to study the growth of cosmic structure, she moved into Earth observation, where she has spent eight years building machine learning and remote sensing pipelines across Australia, Antarctica, and Africa. Caitlin is a member of the Open Data Cube Steering Council, where she volunteers her time supporting the global Earth observation community.



FEATURED PROJECTS

SAR Data Processing and Product Development for Australia and Antarctica | Geoscience Australia

- Led development of processing pipelines converting Level-1 Sentinel-1 SAR data into Level-2 normalised radar backscatter, using open-source tools including ISCE3 and PyroSAR.
- Designed an AWS pipeline in Terraform supporting both ongoing processing and full historical reprocessing of the Sentinel-1 archive.
- Co-developed a Python library for handling digital elevation models, including complex cases at the antimeridian.
- Worked with university and government stakeholders to build prototype products, including automated mapping of land-fast and blue ice, and support certification of the product's STAC metadata against the CEOS-ARD specification.

Digital Earth Africa Waterbodies | Digital Earth Africa

- Led adaptation of Digital Earth Australia's Waterbodies pipeline for continental-scale application across Africa, improving computational efficiency and reducing AWS processing costs.
- Designed an Argo Workflows architecture to process 30+ years of Landsat-derived Water Observations from Space data into time series, with automated weekly updates.
- Worked with Digital Earth Africa staff to understand user needs and co-design a computationally efficient mapping approach, now a core program service.

Tree Canopy Mapping | Victorian Department of Environment, Land, Water and Planning

- Led technical development of a suite of tree-cover products, applying deep learning (convolutional neural networks) to 10–20cm aerial RGB imagery across Victoria.
- Used AWS Batch and Docker to process 37 terabytes of aerial imagery stored in AWS S3, building a scalable, cloud-native workflow.
- Developed educational resources with Digital Earth Australia and Digital Earth Africa to help end users access and apply Earth observation data.

LANGUAGES

- Python
- C++

PROFESSIONAL AFFILIATIONS

- Chair, Documentation Subcommittee | Open Data Cube Steering Council | 2024 – Present (member since 2023)
- Steering Council Member | Open Data Cube | 2019 – Present (Chair, 2022–2023)

AWARDS

- Early-Career Contribution to the Australian Earth Observation Community Award | Earth Observation Australia | 2022
- Leading Women in Machine Learning for Earth Observation | Radiant Earth | 2022



CAREER HISTORY

Senior Data Scientist | FrontierSI

- Lead development of continental-scale remote sensing products from Landsat and Sentinel-2 optical imagery, including adapting Digital Earth Australia's Waterbodies pipeline for Digital Earth Africa.
- Improve computational efficiency of processing workflows using Argo, reducing processing time and AWS cloud costs.
- Provide Agile project management and stakeholder communication across large-scale product development projects.
- Consult to other organisations on developing Earth observation products, and support the professional development of junior Data Scientists.

Data Scientist | FrontierSI

- Lead technical development of a suite of tree-mapping products for the Victorian Department of Environment, Land, Water and Planning, applying machine learning to aerial imagery on AWS.
- Apply machine learning to Earth observation data to develop informative, decision-ready datasets.
- Develop educational resources with Digital Earth Australia and Digital Earth Africa to support end users in accessing and applying Earth observation data.

Data Analyst and Modeller | Mezo Research

- Develop analytic tools and statistical/mathematical models for natural and human resource management projects, including clients such as Victoria Police and the Victorian Fisheries Authority.
- Communicate modelling approaches and findings directly to clients to inform management decisions.
- Oversee the company's Agile development process through Git and Jira.



PUBLICATIONS

Adams, C. & Blake, C. (2017). Improving constraints on the growth rate of structure by modelling the density-velocity cross-correlation in the 6dF Galaxy Survey. Monthly Notices of the Royal Astronomical Society. This paper is available online at arxiv.org/abs/1706.05205

Adams, C. & Blake, C. (2020). Joint growth-rate measurements from redshift-space distortions and peculiar velocities in the 6dF Galaxy Survey. Monthly Notices of the Royal Astronomical Society. This paper is available online at arxiv.org/abs/2004.06399

Bellstedt, S., Forbes, D., Romanowsky, A., Remus, R., Stevens, A., Brodie, J., Poci, A., Mc-Dermid, R., Alabi, A., Chevalier, L., Adams, C., Ferré-Mateu, A., Wasserman, A., Pandya, V. (2018). The SLUGGS Survey: a comparison of total-mass profiles of early-type galaxies from observations and cosmological simulations, to ~4 effective radii. Monthly Notices of the Royal Astronomical Society. This paper is available online at arxiv.org/abs/1803.02373